

FEATURES

- Aluminum or stainless steel body
- Tension and Compression
- Heavy duty
- Sealed version optional
- Easy to customize threads

APPLICATIONS

- Dynamic strain cylinder regulation
- Miniature press-fit device
- Laboratory
- Robotics regulation
- Small size actuators

XFTC321

Miniature Load Cell

SPECIFICATIONS

- Range from 0-500N to 0-10kN
[0-112.4 lbf to 2.2481 klb]
- Tension and/or Compression
- Threaded Female Mechanical Fitting

The **XFTC321** series has been specifically developed to measure tension and compression in static and dynamic applications. The miniature size facilitates testing where space is at a premium.

The sensing element is fitted with a fully temperature compensated Wheatstone bridge equipped with high stability micro-machined silicon strain gages. The use of silicon strain gages optimizes the load cell's performance at low ranges and frequencies.

A strain relief spring strengthens the cable output. With two female threads, the **XFTC321** is easily installed in industrial or OEM applications.

On request, Instruction documents can be provided to ease the selection and use of our sensors and provide helpful tips.

PERFORMANCE SPECIFICATIONS (typical values at temperature 23°C)

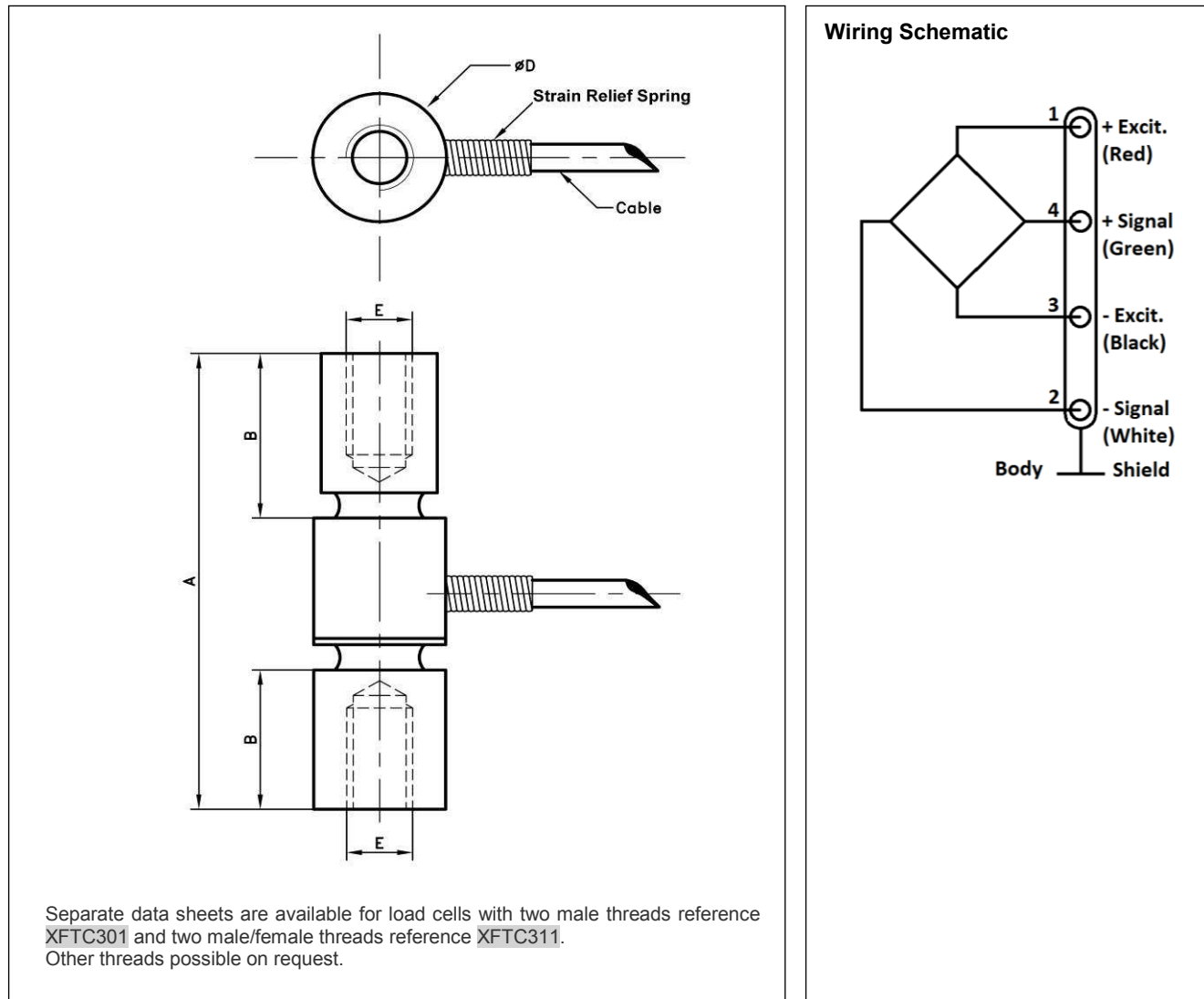
Ranges (FS) (N)	500 N	1 kN	2 kN	5 kN	10 kN
Ranges (lbf)	112.4	224.8	449.6	1124.0	2248.1
Material	Aluminium		Stainless steel		
Stiffness (N/m)	1.5E+08	1.5E+08	3.7E+08	4.6E+08	7.6E+08
Stiffness (lbf/ft)	1.0E+07	1.0E+07	2.5E+07	3.2E+07	5.2E+07
Sensitivity (FSO)	70 mV	150 mV	100 mV	150 mV	

Specifications	
Power supply	10Vdc
Overrange Without Damage	2x FS
Overrange Without Destruction	3x FS
Offset	<±10 mV
Input Impedance	1500 ohms
Output Impedance	500 ohms
Linearity	< ±0.5% FS
Hysteresis	< ±0.5% FS
Operating Temperature Range (OTR)	-40°C to +120°C (-40 to 250°F)
Compensated Temperature Range (CTR)	0°C to +60°C (32 to 140°F)
Thermal Zero Shift in CTR	< 2% FS/50°C
Thermal Sensitivity Shift in CTR	< 2%/50°C
Insulation	> 100 Mohms
Protection Index	IP50

Notes

1. Signal goes positive in tension with standard wiring configuration. Other signal output on request
2. Shielded cable with 4 wires (AWG36/28), standard length 2 m [6.5 ft] with strain relief spring
3. Material: Body in stainless steel or aluminum alloy depending on F.S.
4. CE conformance according to EN 61010-1, EN 50081-1, EN 50082-1

DIMENSIONS & WIRING SCHEMATIC (IN METRIC AND IMPERIAL)

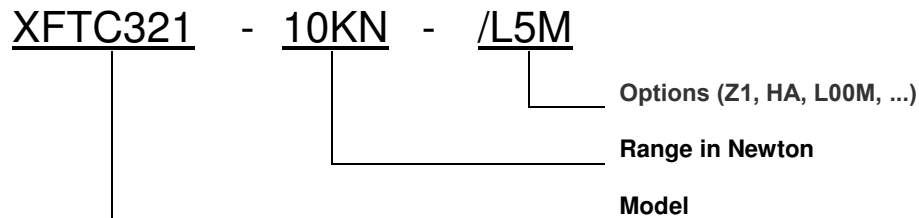


Dimensions in mm [inch]

Full Scale Range in N	500-1k	2k	5k-10k
A	36 [1.42]		46 [1.81]
B	11 [0.43]		13 [0.51]
C	12.5 [0.49]		14 [0.55]
Ø D	10 [0.39]		16 [0.63]
E Thread	M5		M10
Internal depth	8 [0.31]		10 [0.39]

OPTIONS

Z0	CTR -40°C to +20°C (-40°F to 68°F)
Z04	CTR -40°C to +90°C (-40°F to 194°F)
Z1	CTR -20°C to +40°C (-4°F to 104°F)
Z3	CTR +20°C to +80°C (68°F to 176°F)
Z35	CTR +20°C to +120°C (68°F to 248°F)
Z36	CTR +20°C à +150°C (68°F to 302°F) - Only available for range 2kN / 5kN / 10kN
HA	Accuracy (CNL&H) $\leq \pm 0.5\%$ F.S.
L5M , L10M , L15M	Special cable length (ex : L5M = 5m length) (Standard cable length 2m)

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